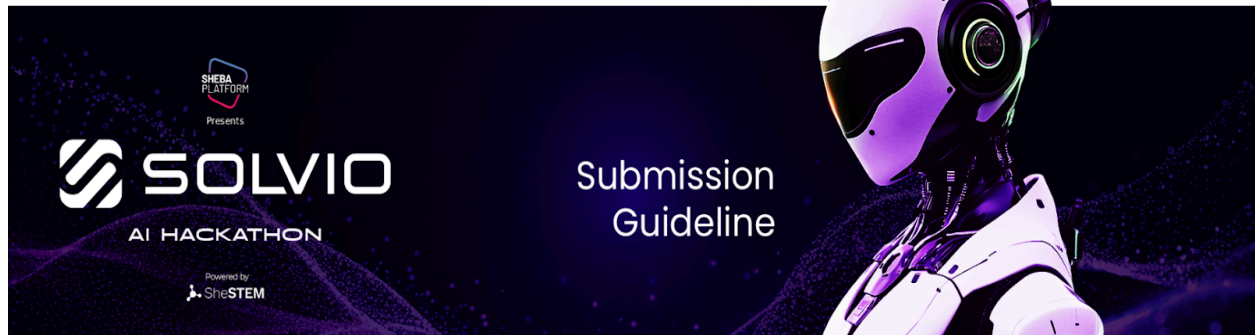




Project Proposal: ServiSphere - The AI-Powered Trust & Intelligence Network for Services

Team Name: AI_Builder

Challenge Category: Online Service Platform Industry

Date: October 28, 2025

1. Team Details

Team Name: AI_Builder

Members:

- **Md. Mahfujul Karim Sheikh** – Northern University of Business and Technology Khulna – Dept. of CSE (GitHub: [Mahfujul-01726](#))
- **Md. Sahadat Hossen Sajib** – Varendra University – Dept. of CSE (GitHub: [sahadat531](#))
- **Maria** – Shahjalal University of Science and Technology – Dept. of Software Engineering (GitHub: [maria2021831011](#))

- **Jasia Hasan Mumu** – Northern University of Business and Technology Khulna – Dept. of CSE

Team Summary: A dynamic AI innovation team with diverse expertise in machine learning, software engineering, and data-driven system design. We are passionate about developing scalable, ethical, and impactful AI applications that solve real-world problems.

Core Idea: The "Trust-First" Paradigm Shift

ServiSphere is not just another service marketplace. It is a decentralized AI network that shifts the paradigm from simple listing-and-booking to verifiable trust and intelligent matching. We solve the core problem of authenticity by building a portable, AI-verified digital identity for every service provider and a

smart, contextual discovery engine for every user.
Core Philosophy: “Trust is not a review score; it is a verifiable data graph.”

Problem Analysis: Beyond Overwhelming Choices

The current service platform economy is broken due to information asymmetry and a crisis of trust.

For Users:

- **Fake Reviews & Inflated Ratings:** Hard to distinguish authentic feedback.
- **One-Size-Fits-All Search:** Keyword-based results fail to capture context.
- **Lack of Verifiable Proof:** No evidence of skills or past work quality.

For Service Providers:

- **High Platform Commissions:** Earnings reduced by marketplace fees.
- **Unfair Competition:** Honest providers overshadowed by manipulated reviews.
- **No Data Ownership:** Reputation locked to one platform, not transferable.

Proposed Solution: The ServiSphere Ecosystem

Our solution is a **three-pillar ecosystem powered by AI.**

Pillar 1: TrustGraph ID™ (Decentralized Digital Identity)

Each provider gets a unique, portable TrustGraph ID containing:

- **Verified Skill Badges:** Certificates validated via document AI.
- **Work Portfolio:** AI-analyzed images/videos for quality assurance.
- **Dynamic Trust Score:** Based on sentiment analysis, repeat customers, on-time completion, and complaint resolution.

Pillar 2: Context-Aware Discovery Engine

Goes beyond keywords using:

- **BERT-based NLP:** Understands intent (“fix a leak” vs. “install faucet”).
- **Contextual Filters:** Time, urgency, user behavior, and weather conditions.

- **Multi-Modal Search:** Users can speak (Bangla/English) or upload problem images for AI-based recommendations.

Pillar 3: ServiBot – The Smart Matchmaker & Scheduler

Acts as an AI intermediary that:

- Handles user queries, negotiates time using live traffic data, and sends reminders.
- Provides providers with a **Demand Heatmap** predicting high-demand zones.

Key Innovations & Uniqueness		
Feature	Existing Platforms	ServiSphere’s Advantage
Trust Mechanism	Static, user-written reviews	TrustGraph ID™ : AI-verified, portable, multi-source trust
Search & Discovery	Keyword-based for all	Context-Aware AI : Personalized, multimodal, intent-based
Provider Empowerment	Basic listing, high fees	Data Ownership & AI Insights for business growth
Business Model	High transaction fees	Low-Fee + SaaS Model for fairness
Accessibility	English text-only	Bilingual Voice & Text (Bangla + English)

Feasibility & Growth Potential

Technical Feasibility: Uses proven technologies (TensorFlow, PyTorch, Neo4j, Flutter). Models for sentiment, vision, and NLP are fine-tuned for the service domain.

Development Roadmap:

- **Phase 1 (Months 1–6):** MVP with booking, profiles, TrustGraph basics. Pilot in Dhaka & Khulna.
- **Phase 2 (Months 7–12):** Add image/voice AI & demand heatmap. Target 30k users.
- **Phase 3 (Year 2):** Premium SaaS for providers. Expand to other cities. 300k+ users.

- **Phase 4 (Year 3+):** Open API for TrustGraph integration globally.

Technology Stack Recommendation

Layer	Technology
Frontend	Flutter (cross-platform)
Backend & API	Node.js + FastAPI
Database	PostgreSQL (structured) + Neo4j (TrustGraph)
AI/ML Core	Python, TensorFlow/PyTorch, Scikit-learn
NLP & Voice	BERT, BanglaBERT, Google Speech APIs
Computer Vision	OpenCV, TensorFlow Lite (on-device)
Cloud & Deployment	Google Cloud / AWS
Security	AES-256, OAuth 2.0, Federated Learning

Projected Impact

User Reach:

- **Year 1:** 30,000 pilot users.
- **Years 2–3:** 300,000+ across major cities.
- **Industry & Social Impact:**
- 40% fewer service disputes via verified profiles.
- Empowerment of SMEs & freelancers through portable, verifiable reputation.
- 25% higher booking confidence and user satisfaction.

- Integration with bKash/Nagad for financial inclusion.
Scalability: Easily adaptable to international service industries.

Conclusion

ServiSphere solves the trust and authenticity gap in online service markets by introducing AI-verifiable digital identities and contextual discovery. We empower users with confidence and providers with fairness. With strong technical expertise and cross-domain collaboration, **Team AI_Builder** is ready to transform Bangladesh's service economy into a transparent, data-driven ecosystem.



Presents



AI HACKATHON
2025

THE HACKERMAN DAY:

GRAND SHOWDOWN VENUE:
SHEBA HQ, JASHORE IT PARK

REGISTRATION DEADLINE:
28TH OCT, 2025



NO
REG
FEE!

TO REGISTER SCAN QR

Prize Pool
5,00,000 BDT

Cash Price

Direct Recruitment

Trophies

Free Courses

Certificates

Media Feature

Powered by



Kingdom of the Netherlands

Consortium Partners



DEVLEARN

LightCastle

10 MINUTE
SCHOOL

POLICY EXCHANGE

Collaboration Partner
Programming Hero